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United States Patent	12397973
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Darst; Daniel et al.

Pill container and methods

Abstract

A child resistant pill container includes a locking feature, which can be unlocked using a knob. The knob can be a rotatable knob or a slide knob. The knob will be fixed or locked in place until being released/unlocked by squeezing an actuator. The squeeze action allows the knob to move out of its fixed position and move by either rotation or linearly sliding. The motion of the knob moves a lock bar from its locked position to an unlocked position. While the lock bar is in the unlocked position, one or more of the lids can be unlatched and lifted from their closed position to an open position. When the knob is released, a spring action returns the knob back to the fixed (locked) position, moving the lock bar back into the locked position, and any lids still in the closed position are again locked in place.

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Appl. No.: 18/668962

Filed: May 20, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240359882 A1	Oct. 31, 2024

Related U.S. Application Data

continuation parent-doc US 17496450 20211007 US 11987429 child-doc US 18668962
us-provisional-application US 63089623 20201009

Publication Classification

Int. Cl.: **B65D50/00** (20060101); **A61J1/03** (20230101); **B65D25/04** (20060101); **B65D43/16** (20060101); **B65D50/04** (20060101); **B65D55/10** (20060101)

U.S. Cl.:

CPC **B65D50/045** (20130101); **A61J1/03** (20130101); **B65D25/04** (20130101); **B65D43/16** (20130101); **B65D55/10** (20130101); B65D2215/02 (20130101)

Field of Classification Search

CPC: A61J (1/03); B65D (25/04); B65D (43/16); B65D (50/045); B65D (55/10); B65D (2215/02)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/496,450, filed Oct. 7, 2021, which claims the benefit of U.S. Provisional Patent Application No. 63/089,623, filed Oct. 9, 2020; the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) This disclosure relates generally to containers with hinged mating lids, often used for holding medicine, such as pills. In particular, this disclosure relates to containers having child resistant features and methods of opening such containers.

BACKGROUND

(2) Containers with hinged mating lids are well known. Some containers can include multiple compartments. See, for example, commonly assigned U.S. Pat. No. 7,624,890, incorporated herein by reference.

(3) When these containers hold medicine or vitamins, it is helpful to have features that will not allow a child to easily access the contents, for safety reasons. While there are pill containers that are considered child-deterrent, improvements are desirable for containers that are child resistant per the standard of US 16 CFR 1700.15.

SUMMARY

(4) In general, this disclosure is directed to pill containers that are an improvement over the prior art.

(5) In general, the improved pill containers will do at least one, and preferably all, of the following functions: (i) prevent children from accessing the contents of compartments in the pill container; (ii) after opening one container by releasing a lid, automatically locking all non-open lids; and (iii) to close any open lid, requiring an actuator to be moved into an “unlock position” first.

(6) A container includes: (a) a base having surrounding wall defining a cavity with an interior volume; (b) a lid pivotably attached to the base and sized to removably cover the cavity so that the lid is movable between a closed position covering the cavity and an open position exposing the cavity; (c) a locking mechanism to control locking and unlocking of the lid to the base; and (d) an actuator to move the locking mechanism from the locked position to unlocked position.

(7) In preferred implementations, the container includes (a) a plurality of subcavities within the base, and a plurality of lids; one lid for each of the subcavities; (b) the locking mechanism controls locking and unlocking of all of the lids to the base; and (c) the actuator biasing the locking mechanism into the locked position, such that when released, the actuator locks all lids that are covering their respective subcavity.

(8) Preferably, the actuator is constructed and arranged such that, to move one of the lids from the open position to the closed position, the actuator must be moved against the bias into the unlocking position.

(9) In example embodiments, the lid has a first member of a hook and catch arrangement to selectively secure the lid to the base, when the lid is in the closed position, and selectively release the lid from the base; and (ii) the base has a second member of a hook and catch arrangement to selectively secure the lid to the base, when the lid is in the closed position, and selectively release the lid from the base.

(10) In many embodiments, a lock bar is slidably engaging with the second member of the hook and catch arrangement; the lock bar being movable between a locked position locking the lid to a secure position to the base, and an unlocked position releasing the lid from the base.

(11) Preferably, the actuator comprises a knob having a squeeze actuator to release the knob from a fixed position to allow the knob to move and push on the lock bar to urge the lock bar from the locked position to unlocked position.

(12) In some examples, the knob moves rotationally after released from the fixed position.

(13) In some examples, the knob slides linearly after released from the fixed position.

- (14) The knob may include a spring that returns the knob to the fixed position.
- (15) Many implementations include the spring as a torsion spring.
- (16) Preferably, the knob includes a spring-retaining aperture constructed and arranged to receive a portion of the torsion spring and retain the torsion spring in place.
- (17) Some embodiments include the container includes a plurality of subcavities within the base, and a plurality of lids; one lid for each of the subcavities.
- (18) In some embodiments, the base and lid are each shaped to be a partial cylinder.
- (19) In one or more embodiments, the knob includes an internal ramp; and the lock bar engages against the ramp as the knob is moved from the fixed position resulting in the lock bar sliding in a direction parallel to a longitudinal axis of the base to the unlocked position.
- (20) The lock bar can include at least one notch corresponding to each hook and catch arrangement; and when the lock bar is in the locked position, the notch is engaged with a projection on one of the first or second members of the hook and catch arrangement, and when the lock bar is in the unlocked position, the notch is free of engagement with the projection.
- (21) In examples, the squeeze actuator includes a pair of opposite outwardly facing knurled surfaces; and a pair of inwardly extending ribs.
- (22) In another aspect, a method of opening a container locked closed with a lock bar is provided. The method comprising: (a) grasping a knob and moving the knob from a fixed position to a released position by squeezing a squeeze actuator on the knob; (b) moving the knob to move the lock bar from a locked position to an unlocked position; and (c) while still grasping the knob, moving a lid from a closed position to an open position to expose an open cavity in the container.
- (23) Preferably, the step of moving a lid includes lifting the lid to release engagement of a secondary latch arrangement between the lid and a base of the container.
- (24) In some examples, the step of moving the knob includes rotating the knob.
- (25) In some examples, the step of moving the knob includes linearly moving the knob.
- (26) Some methods having the step of linearly moving the knob will include sliding the knob relative to a sidewall of the container.
- (27) Some methods further comprise releasing the knob and permitting the knob to spring back to the fixed position, and permitting the lock bar to spring back to the locked position.
- (28) Preferably, the container includes a plurality of subcavities and a plurality of lids; one lid for each of the subcavities; and the step of moving the knob to move the lock bar from a locked position to an unlocked position controls locking and unlocking of all of the lids to the base.
- (29) The method can include moving the lid from the open position to the closed position by moving the knob to move the lock bar from the locked position to the unlocked position; and while still grasping the knob, moving the lid from the open position to the closed position.
- (30) A variety of examples of desirable features or methods are set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practicing various aspects of the disclosure. The aspects of the disclosure may relate to individual features as well as combinations of features. It is to be understood that both the foregoing general description and the following detailed description are explanatory only, and are not restrictive of the claimed invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a top perspective view of an embodiment of a pill container in the closed position;
- (2) FIG. 2 is an exploded perspective view of the pill container of FIG. 1;
- (3) FIG. 3 is a top plan view of the pill container of FIG. 1;
- (4) FIG. 4 is a cross-sectional view of the pill container of FIG. 1, and showing the lock bar in an

unlocked position, the cross section being taken along the line A-A of FIG. 3;

(5) FIG. 5 is an enlarged view of a portion shown at B of FIG. 4;

(6) FIG. 6 is a view similar to FIG. 4, but showing the lock bar in a locked position;

(7) FIG. 7 is an enlarged view of the portion shown at B of FIG. 6;

(8) FIG. 8 is a top plan view of the pill container of FIG. 1, and showing the knob rotated from a fixed position of FIG. 3 to an unlocking position;

(9) FIG. 9 is a cross-sectional view of the pill container, the cross section being taken along the line A-A of FIG. 8;

(10) FIG. 10 is a perspective view showing engagement between the knob and the lock bar of the pill container of FIG. 1;

(11) FIG. 11 is a perspective view of a second embodiment of a pill container, the pill container being shown with all the lids shown in a closed position;

(12) FIG. 12 is an exploded perspective view of the container of FIG. 11;

(13) FIG. 13 is a top plan view of the container of FIG. 11;

(14) FIG. 14 is a cross-sectional view of the container of FIG. 11 showing the lock bar in an unlocked position, the cross section being taken along the line A-A of FIG. 13;

(15) FIG. 15 is an enlarged view of section B of FIG. 14;

(16) FIG. 16 is a cross-sectional view similar to that shown in FIG. 14, but showing the lock bar in a locked position;

(17) FIG. 17 is an enlarged view of section B of FIG. 16;

(18) FIG. 18 is a side view of the container of FIG. 11;

(19) FIG. 19 is a cross-sectional view of the container, the cross section being taken along the line 19-19 of FIG. 18;

(20) FIG. 20 is an enlarged view of the portion at section A of FIG. 19;

(21) FIG. 21 is an enlarged view of portion B shown at FIG. 19;

(22) FIG. 22 is a side view of the container of FIG. 11;

(23) FIG. 23 is a cross-sectional view of the container of FIG. 11, the cross section being taken along the line 23-23 of FIG. 22;

(24) FIG. 24 is an enlarged view of section A of FIG. 23;

(25) FIG. 25 is an enlarged view of section B of FIG. 23;

(26) FIG. 26 is a perspective view showing engagement between the lock bar and knob used in the container of FIG. 11;

(27) FIG. 27 is an enlarged view of section A of FIG. 26;

(28) FIG. 28 is a view similar to FIG. 26, but showing the lock bar being moved to a locked position;

(29) FIG. 29 is an enlarged view of section A of FIG. 28;

(30) FIG. 30 is a perspective view of a third embodiment of a pill container, the lids being in a closed position;

(31) FIG. 31 is an exploded perspective view of the container of FIG. 30;

(32) FIG. 32 is a top plan view of the container of FIG. 30;

(33) FIG. 33 is a cross-sectional view of the container of FIG. 30, the cross section being taken along the line A-A of FIG. 32, the lock bar being in an unlocked position;

(34) FIG. 34 is an enlarged view of section A of FIG. 33;

(35) FIG. 35 is a cross-sectional view similar to the view shown in FIG. 33, but showing the lock bar in a locked position;

(36) FIG. 36 is an enlarged view of section A of FIG. 35;

(37) FIG. 37 is a top plan view of the container of FIG. 30;

(38) FIG. 38 is cross-sectional view taken along the line 38-38 of FIG. 37;

(39) FIG. 39 is a perspective view showing the lock bar and knob of the container of FIG. 30;

(40) FIG. 40 is an enlarged view of section A of FIG. 39;

- (41) FIG. 41 is a perspective view of a fourth embodiment of a container, constructed in accordance with principles of this disclosure;
- (42) FIG. 42 is an exploded perspective view of the container of FIG. 41;
- (43) FIG. 43 is a front view of the container of FIG. 41;
- (44) FIG. 44 is a perspective view of the knob and lock bar of the container of FIG. 41;
- (45) FIG. 45 is a cross-sectional view taken along the line A-A of FIG. 43 and showing the locked bar in a locked position;
- (46) FIG. 46 is an enlarged view of section A of FIG. 45;
- (47) FIG. 47 is a cross-sectional view similar to FIG. 45 and showing the lock bar in an unlocked position;
- (48) FIG. 48 is an enlarged view of section A of FIG. 47;
- (49) FIG. 49 is a perspective view showing the container of FIG. 41, with the lid removed and showing a separate container stored therein;
- (50) FIG. 50 is another perspective view of FIG. 49;
- (51) FIG. 51 is a perspective view of a portion of the containers of FIGS. 1-40, showing the lids in an open position;
- (52) FIG. 52 is an enlarged view showing a cross-section, representative of the containers of FIGS. 1-50, with a lid in the closed position with a secondary latch arrangement engaged, and a lock bar in an unlocked position;
- (53) FIG. 53 is a bottom plan view of the containers of FIGS. 1-40;
- (54) FIG. 54 is a top perspective view of another embodiment of a pill container in the closed position;
- (55) FIG. 55 is a front view of the pill container of FIG. 54;
- (56) FIG. 56 is a top plan view of the pill container of FIG. 54;
- (57) FIG. 57 is an exploded, perspective view of the pill container of FIG. 54;
- (58) FIG. 58 is a perspective view of an enlarged portion of an end of the pill container of FIG. 57;
- (59) FIG. 59 is a perspective view of a knob used with the pill container of FIG. 54;
- (60) FIG. 60 is a perspective view of the pill container of FIG. 54 with all the lids shown in an open position, and a front panel moved from the container;
- (61) FIG. 61 is a perspective view of an enlarged portion of an end of the pill container of FIG. 60; and
- (62) FIG. 62 is a perspective view of an enlarged portion of bottom, end of the pill container of FIG. 60.

DETAILED DESCRIPTION

(63) Pill containers described herein have a locking mechanism that meets a child-resistant protocol, as specified in Federal Regulations, such as 16 CFR 1700.15. The pill containers have a secure closure to avoid compartments opening or spilling when dropped. They are easy to open for those with dexterity issues (Ease of Use from Arthritis Foundation). They are made from food-grade, BPA plastic. They last through multiple uses without tabs breaking off, compartments not closing, etc.

(64) In general, the pill containers described herein will do at least one, and preferably all, of the following functions: (i) prevent children from accessing the contents of compartments in the pill container; (ii) after opening one container by releasing a lid, automatically locking all non-open lids; and (iii) to close any open lid, requiring an actuator to be moved into an “unlock position” first.

(65) In preferred non-limiting examples, the pill container examples disclosed herein include a locking mechanism (e.g., a lock bar), which either locks or releases one or more lids of the container. An actuator (e.g., a knob) is actuated to move the lock bar from the locked position to an unlocked position, releasing one or more of the lids. The actuator preferably automatically locks all unopened containers, and/or it preferably needs to be moved to the “release” or “unlock” position

to re-close any open containers.

(66) FIGS. **1-10** are directed to a first embodiment, in which to open the lids, the knob is squeezed and rotated, and the lids can be opened independently by lifting each lid to release each lid from a secondary latch arrangement. There is a built in return spring so that when the knob is released, the knob will return to a fixed position, locking all remaining closed lids. The embodiment of FIGS. **11-29** is similar to embodiment of FIGS. **1-10**, but do not use a torsion spring as the return spring. The embodiment of FIGS. **30-40** includes the knob being moved linearly instead of rotated. The embodiment of FIGS. **41-50** is a similar concept, in that the end knob is squeezed and rotated, which will allow the enclosure to open. The enclosure of FIGS. **41-50** can be used to store a pill box, or round pill boxes, or any of a variety of items. The embodiment of FIGS. **54-62** is similar to the embodiment of FIGS. **1-10** and includes some additional features for added strength and functionality.

(67) A pill container in accordance with principles of this disclosure is shown in general at **100**. The container **100** can be used for the convenient storage of a variety of materials, including pills, medicines, vitamins, and the like. The container **100** can also be used to store any other small items such as buttons, screws, fasteners, sequins, or other various materials.

(68) An example embodiment, representative of each of the embodiments of FIGS. **1-40**, is shown in FIGS. **1, 11, 30, 41, 51**, and **54**. It should be noted that FIG. **51** shows a representative container **100**, but with only two cavities. It should be understood that in FIG. **51**, there can be 7 or more cavities, like that shown in FIGS. **1** and **11**.

(69) Example Base

(70) The container **100** includes a base **102**. The base **102** has a surrounding wall **104** which defines a cavity **106** (FIG. **51, 60**) with an interior volume **108**. The cavity **106** is for holding the contents of the container **100**.

(71) While many different configurations are possible, in the example shown, the surrounding wall **104**, in this embodiment, includes a front wall **128**, an opposite back wall **130**, a first side wall **132**, and a second side wall **134** opposing the first side wall **132**. The first and second side walls **132, 134** extend between the front wall **128** and back wall **130**. A bottom **124** (FIG. **53**) forms the bottom surface of the base **102**.

(72) In FIG. **53**, a bottom plan view of the container **100** is shown, and the bottom **124** is visible. The bottom **124** includes a plurality of feet **222** extending therefrom. The feet **222** will preferably be covered by a friction-enhancing material, such as silicone pads. In this case, there are 4 feet **222**, adjacent to each corner of the container **100**. In use, the container **100** will rest on the feet **222**, with the silicone pads providing slip resistance. As will be described further below, to open the container **100**, two hands are used, so it is helpful to have the container **100** to have the slip-resistant feet **222**.

(73) In the embodiment of FIGS. **41-50**, the container **100** can either be used empty, or it can hold a pill container of any shape, including a cylinder shaped pill container **190**. One or more cylinder shaped containers **190** can be stored within the cavity **106** of the container **100**.

(74) Example Lid

(75) In accordance with principles of this disclosure, the container **100** further includes a lid **162**. The lid **162** is sized to removably cover the cavity **106** over the mouth **126** so that the lid **162** is movable between a closed position covering the cavity **106** (FIG. **1**) and an open position (FIG. **51**) exposing the cavity **106**. The lid **162** has an interior surface **164**, which faces the cavity **106**, when the lid **162** is in the closed position. The lid **162** has an opposite exterior surface **166** (FIG. **1**), which always faces away from the cavity **16**.

(76) Still in reference to FIG. **51**, the lid **162** is secured to the back wall **130** of the base **102**. While many examples are possible, in the one shown, the lid **162** is pivotably secured to the back wall **130** of the base **102**. In this example, the lid **162** is secured to the back wall **130** along a hinge line or hinge axis **168** so that the lid **162** can pivot between a closed position (covering the cavity **106**)

and an open position (allowing access to the interior volume **108** of the cavity **106**). The hinge axis **168** is generally parallel to a longitudinal axis **174** (FIG. 3, **56**) of the container **100**. The longitudinal axis **174** passes through both the first side wall **132** and second side wall **134**.
(77) In the embodiments of FIGS. **1-10**, **11-29**, **30-40**, and **54-62** the container **100** includes a plurality of sub-cavities **110** (FIG. **51**) within the base **102**. Each sub-cavity **110** functions as an individual compartment to hold pills, for example. A plurality of lids **162** are provided, with one lid **162** for each of the sub-cavities **110**. In the embodiment of FIGS. **41-50**, the base **102** and lid **162** are each shaped to be a partial cylinder, such as roughly a half cylinder. The sub-cavities **110** (FIG. **51**) are smoothly contoured and shaped to allow for ease of scooping out pills. In the embodiment of FIGS. **1-10**, there are 7 sub-cavities **110**, while in the embodiment of FIGS. **54-62**, there are 4 sub-cavities **110**. It should be understood there can be fewer than 4 and more than 7, in other embodiments.

(78) In FIG. **60**, the lids **162** include ribs **330** along an outer periphery of the interior surface **164** to help with strength and rigidity.

(79) Example Hook and Catch Arrangement

(80) The lid **62** and the base **102** include a hook and catch arrangement **182**, shown in FIG. **51**. The FIG. **51** embodiment shows only two cavities **106**, but is representative of each of the containers **100** shown in FIGS. **1-40**. In FIG. **51**, the lid **162** includes a first member **184** of the hook and catch arrangement **182**, while the base **102** has a second member **186** of the hook and catch arrangement **182**. The hook and catch arrangement **182** allows for selective locking of the lid **162** to the base **102**, and selective unlocking of the lid **162** from the base **102**. When the lid **162** is unlocked from the base **102**, the lid **162** can be lifted from the base **102** to move the lid **162** from the closed position to the open position.

(81) Still in reference to FIG. **51**, the hook and catch arrangement **182** includes a hook and a catch. The hook and catch can be on either the lid **162** or the base **102**. In the particular example shown, the first member **184** of the hook and catch arrangement **182** is a hook **194** projecting from a free end **196** of the lid **162**. The second member **186** of the hook and catch arrangement **182** is a catch **198** within the base **102** and adjacent to the front wall **128** of the surrounding wall **104** the base **102**.

(82) When the lid **162** is in the closed position of FIG. **1**, the hook **194** is located within the catch **198** to lock or hold the lid **162** in place.

(83) Still in reference to FIG. **51**, there is a projection **199** extending from one of the first or second members **184**, **186** of the hook and catch arrangement **182**. The projection **199** interacts with a lock bar **140**, as explained further below. In the example embodiment shown, the projection **199** extends from the hook **194**.

(84) In FIGS. **60** and **61**, the first member **184** further includes a keeper **340** projecting along a plane perpendicular to the plane containing the hook **194**. When the lid **162** is in a closed position, the keeper **340** extends outwardly away from and perpendicular to the front wall **128**. The keeper **340** helps with retention of the lids **162** in a closed position to prevent the lids **162** from popping open or up.

(85) Example Locking Mechanism

(86) The container **100** includes a locking mechanism to control locking and unlocking of the lid **162** to the base **102**.

(87) In non-limiting examples, the locking mechanism comprises a lock bar **140**. The lock bar **140** is slidably engaging with the second member **186** of the hook and catch arrangement **182**. The lock bar **140** is movable between a locked position, which locks the lid **162** to a secure position to the base **102** and an unlocked position releasing the lid **162** from the base **102**. In FIG. **2**, only a partial section of the lock bar **140** is shown. In other embodiments, the full lock bar **140** is illustrated.

(88) As can be seen in FIG. **2**, the lock bar **140** has at least one notch **142**. There is at least one notch **142** which corresponds to each hook and catch arrangement **182**. When the lock bar **140** is in

the locked position, the notch **142** is engaged with the projection **199** (FIGS. **51** and **52**) of the hook and catch arrangement **182**. When the lock bar **140** is in the unlocked position (FIG. **52**), the notch **142** is free of engagement with the projection **199**. As such, when the lock bar **140** is moved from the locked position to the unlocked position, one or more of the lids **162** are released from being locked in the closed position and can be lifted from the closed position to the open position. (89) For the embodiment of FIGS. **1-10**, FIG. **5** shows the unlocked position of the lock bar **140**, while FIG. **7** shows the locked position.

(90) In the embodiment of FIGS. **11-29**, FIG. **15** shows the unlocked position of the lock bar **140**, while FIG. **17** shows the locked position. In the closed position covering the cavities **106**, the lids **162** are locked in FIG. **17** and unlocked in FIG. **15**.

(91) In the embodiment of FIGS. **30-40**, FIG. **34** shows the locked position of the lock bar **140**, while FIG. **36** shows the unlocked position.

(92) In the embodiment of FIGS. **41-50**, FIG. **46** shows the locked position of the lock bar **140**, while FIG. **48** shows the unlocked position. The lid **162** is shown removed from the base **102**, but if it were in place in the closed position, FIG. **46** would show the lid **162** locked in place, while FIG. **48** would show the lid **162** unlocked.

(93) FIG. **62** shows a bottom view of a portion of the container **100**. In this embodiment, a rib **350** projects from the base **102** and acts as a guide to align the lock bar **140**.

(94) In each of these embodiments, the lock bar **140** has an end heel **180** with an extending flange **181**. The flange **181** will press against the side wall **132** of the base **102** and acts as a cantilevered spring that forces the lock bar **140** from the unlocked position back to the locked position. As explained further below, a portion of the lock bar **140** engages against other structure to move the lock bar from the locked position to the unlocked position. In the examples explained below, the end heel **180** of the lock bar slidingly engages against a ramp **172** within a knob **146**, when the knob **146** is moved from its fixed position to a released position.

(95) Example Secondary Latch Arrangement

(96) In reference again to FIG. **51** and now to FIG. **52**, each of the containers **100** includes a secondary latch arrangement **210** (FIG. **52**). The secondary latch arrangement **210** functions to secure each lid **162** in the closed position covering the sub-cavities **110**, such that the lids **162** remain closed after the lock bar **140** is moved to the unlocked position. While many different embodiments are possible, in the example shown, the secondary latch arrangement **210** includes a pair of latch hooks **212**, **213** and a mating pair of latch catches **214**, **215**. The latch hooks **212**, **213** engage or hook into the latch catches **214**, **215**. The latch hooks **212**, **213** can be released from engagement with the latch catches **214**, **215** by applying a small pulling force to separate them. In this embodiment, the latch hooks **212**, **213** project from the free end **196** of the lids **162**, while the latch catches **214**, **215** are in the base **102**, adjacent the front wall **128**. In the example shown, the catch **198** of the hook and catch arrangement **182** is located between the latch catches **214**, **215**. In FIG. **52**, the lock bar **140** is shown in the unlocked position, freeing engagement with the projection **199**.

(97) One of the advantages of the secondary latch arrangement **210** is that when the lock bar **140** moves to the unlocked position, all of the lids **162** do not automatically open. Rather, an additional step taken by the user is needed to lift the selected lid **162** to release engagement of the secondary latch arrangement **210**. This contributes to the child-resistant nature of the container **100**.

(98) Still in reference to FIG. **51**, in the embodiment shown, each of the lids **162** includes a finger tab **218**. The finger tab **218** is shaped for convenient engagement or manipulation by a human finger to lift the lid **162** to disengage the secondary latch arrangement **210** and pivot the lid **162** about the hinge joint **168**.

(99) Example Actuator

(100) The container **100** includes an actuator to move the locking mechanism from the locked position to unlocked position.

(101) In non-limiting examples, the actuator comprises a knob **146**. The knob **146** has a squeeze actuator **148** that releases the knob **146** from a fixed position to a position which allows the knob **146** to move and push on the lock bar **140**, which urges the lock bar **140** from the locked position to the unlocked position.

(102) In the embodiment of FIGS. **1-10**, FIGS. **11-29**, FIGS. **41-50**, and **54-62**, the knob **146** moves rotationally and is in the form of a turn dial **150**.

(103) In the embodiment of FIGS. **30-40**, the knob **146** is in the form of a slide tab **152** and it slides linearly along one of the side walls **132**, **134** of the container **100**.

(104) In general, the squeeze actuator **148** includes a pair of opposite outwardly facing grip or knurled surfaces **154**, **155**. The squeeze actuator **148** also includes a pair of inwardly extending ribs **156**, **157** (not shown in the FIG. **41-50** embodiment, but it should be understood that the knob **146** in the FIG. **41-50** embodiment is analogous to the FIGS. **1-29** embodiments). When the grip or knurled surfaces **154**, **155** are squeezed in a direction toward each other, the ribs **156**, **157** are moved inwardly, toward each other and moved into open channels **158**, **159** (FIGS. **2**, **9**, **12**, **31**, **38**, **61**) formed in the sidewall **132** of the base **102**, which allows the knob **146** to move from its fixed position. In the embodiments that include the knob **148** as the dial **150**, it allows the knob **146** to rotate. In the embodiment of FIGS. **30-40**, when the knob **46** is in the form of slide tab **152**, the knob **146** can move from its fixed position to move linearly along the side of the container **100**.

(105) The knob **146** includes an internal ramp **172** (FIGS. **9**, **10**, **21**, **25**, **27**, **40**, **44**, **59**) projecting axially from an internal surface of the knob **146**. As the knob **146** is moved from its fixed position, the lock bar **140** (for example, the end heel **180** of the lock bar **140**) engages against the ramp **172**, which results in the lock bar **140** sliding in a direction parallel to the longitudinal axis **174** (FIG. **3**) of the base **102** to the unlocked position. When the lock bar **140** is moved to the unlocked position, the flange **181** is moved and flexed against the sidewall **132** and held in compression until the knob **146** is released, in which case the flange **181** returns to its uncompressed position, and slides the lock bar **140** back into the locked position. The flange **181** can be guided within guide posts **181a**, **181b** (FIG. **58**) projecting from the side wall **132**, as the end heel **180** moves against the ramp **172**.

(106) In the embodiments of FIGS. **1-29**, the knob **146** includes an internally projecting central post **147**, which is received within an aperture **149** in the sidewall **132** of the base **102**, to hold the knob **146** in place against the base **102**. The post **147** can include an end flange **147a** (FIGS. **25** and **27**) to secure the knob **146** in place. In FIGS. **57-61**, there can be a projecting receiving post **149a**, extending from the sidewall **132** to receive the post **147**. In FIG. **59**, the post **147** can be in the form of a split post **147**.

(107) In the embodiment of FIG. **30-40**, the knob **146** includes two internally projecting post pairs **231**, **232** (FIG. **40**), which are received within apertures **236**, **237** (FIGS. **31** and **38**) in the sidewall **132**. The post pairs **231**, **232** can also have end flanges **231a**, **232a** to secure the post pairs **231**, **232** in place against the base **102**.

(108) Each of these embodiments includes a mechanism that automatically returns the knob **146** to the fixed position, when the squeeze actuator **148** is released. In the embodiment of FIGS. **1-10**, a spring in the form of a torsion spring **178** (FIGS. **2**, **10**, and **58**) introduces a bias to push against the rotation of the knob **146**. The knob **146** of FIGS. **1-10** includes a spring stop **173**, extending from an internal surface of the knob **146** and forming a backstop for the spring **178**. The knob **146** of FIG. **59** includes a spring-retaining aperture **308** constructed and arranged to receive a portion (e.g., a free end **310**, FIG. **58**) of the torsion spring **178** and retain the torsion spring **178** in place. A hole **312** (FIG. **58**) in the sidewall **132** also engages the spring **178**. In particular, hole **312** is positioned to preload the spring **178** to assist with “auto-locking”, i.e., ensuring that the knob **146** returns to the fixed position, which moves the lock bar **140** from its unlocked position to a locked position.

(109) As mentioned above, the spring **178** ensures that the knob **146** returns to the fixed position, which moves the lock bar **140** from its unlocked position to a locked position. When it is desired to

move one of the lids **162** from the open position to a closed position, the knob **146** must be moved against the bias of the spring **178** from its fixed position to the released, or unlocking, position which allows the knob **146** to move and push on the lock bar **140**. The lock bar is then moved to the unlocked position, and the lid **162** can be closed in a covering position.

(110) Still in reference to FIG. **58**, the side wall **132** further includes spring-return projections **316**, **318**. Preferably, the projections **316**, **318** are molded in with the rest of the base **102**. The projection **316** protrudes into the channel **159**, while the projection **318** protrudes into the channel **158**. The projections **316**, **318** act as a molded in flexible spring to help the spring **178** return to the fixed position, which returns the knob **146** to the fixed position, moving the lock bar **140** from its unlocked position to a locked position.

(111) In the embodiment of FIGS. **11-50**, a molded in flexible tab **179** (FIGS. **27**, **29**, and **38-40**) on the knob **146** acts as spring to force the knob **146** back to the fixed position.

(112) In FIGS. **11-29**, the base **102** has a slot **176** (FIG. **12**) in the sidewall **132**, which receives the tab **179**. When the knob **146** is rotated, the tab **179** flexes within the slot **176**, and acts as a spring to move the knob **146** back to the original, fixed position. This slot **176** engaging with the tab **179** can be implemented in any embodiment.

(113) In FIGS. **30-40**, the base has a flexible tab **177** projecting from the sidewall **132**. The tab **177** interacts with tab **179** (FIGS. **38-40**). When the slide tab **152** is moved, the tabs **177**, **179** engage and flex, such that the slide tab **152** is returned back to the original fixed position.

(114) In the embodiment of FIGS. **41-50**, it should be understood that the knob **146** can have the same features as the other embodiments, and the spring tab **179**/or torsion spring **178** are omitted; also omitted are the ribs **156**, **157**.

(115) Also shown in FIG. **59**, the knob **146** includes, optionally, a plurality of ribs **320** extending/projecting internally. The ribs **320** contribute to strength and rigidity of the knob **146**.

(116) Example Operation

(117) The pill container **100** can be used to: (i) prevent children from accessing the contents of compartments in the pill container **100**; (ii) after opening one container by releasing a lid **162**, automatically locking all non-open lids **162**; and (iii) to close any open lid **162**, requiring the actuator to be moved into an “unlock position” first.

(118) The container **100** can be used as follows: when the container **100** is locked with the lids **162** covering the cavities **106**, the knob **146** is grasped and moved from its fixed (locked) position by using the squeeze actuator **148**. The squeeze actuator **148** is squeezed by pressing the knurled surfaces **154**, **155** toward each other. This squeeze action will allow the knob **146** to move out of its fixed position and move. In the embodiments of FIGS. **1-29** and **41-50**, this motion of the knob **146** is rotation, while in the embodiment of FIGS. **30-40**, the motion of the knob **146** is a sliding linear motion. The motion of the knob **146** moves the lock bar **140** from its locked position to an unlocked position. While the lock bar **140** is in the unlocked position, one or more of the lids **162** can be unlatched and lifted from their closed position to an open position, by applying a pulling (or lifting) force to the selected lid(s) **162** and releasing engagement of the secondary latch arrangement **210** of the selected lid(s) **162**. When the knob **146** is released, a spring action returns the knob **146** back to the fixed (locked) position, moving the lock bar **140** back into the locked position, and any lids **162** still in the closed position are again locked in place. When the lock bar **140** is moved to the unlocked position, the flange **181** presses against the side wall **132** of the base, and when the knob **146** is released, the lock bar **140** moves back to the locked position by the flange **181** acting as a cantilevered spring that forces the lock bar **140** to return to the locked position.

(119) When moving the knob **146** to move the lock bar **140** from a locked position to an unlocked position, this will controls locking and unlocking of all of the lids **162** to the base **102**.

(120) When moving the lid **162** from the open position to the closed position, the knob **146** is actuated to move the lock bar **140** from the locked position to the unlocked position; and then while

still grasping the knob **146**, the lid **162** is moved from the open position to the closed position. The knob **146** is released, and the lid **162** is locked into the closed position.

(121) Among the advantages of these various embodiments, it should be appreciated, as shown in FIG. **51**, that the container **100** can be made in a single, molded piece including the lid **162**, base **102**, pill compartment/cavity **106**/subcavity **110**, and bottom **124** with feet **222**. This one-piece molded design prevents the user from getting access to the locking bar **140**, keeping it safe and child-resistant. The one-piece molded design also contributes to making assembly of the overall container **100** easy, with easy addition of the knob **146** with spring **178/179**.

(122) The above represents example principles. Many embodiments can be made using these principles.

Claims

1. A container comprising: (a) a base having surrounding wall defining a cavity with an interior volume; (b) a lid pivotably attached to the base and sized to removably cover the cavity so that the lid is movable between a closed position covering the cavity and an open position exposing the cavity; (c) a locking mechanism to control locking and unlocking of the lid to the base; and (d) an actuator biased to hold the locking mechanism in a locked position; the actuator being constructed and arranged such that, to move the lid from the open position to the closed position, the actuator is moved against the bias into the unlocking position; (e) the lid has a first member of a hook and catch arrangement to selectively secure the lid to the base, when the lid is in the closed position, and selectively release the lid from the base; (b) the base has a second member of a hook and catch arrangement to selectively secure the lid to the base, when the lid is in the closed position, and selectively release the lid from the base; and wherein the locking mechanism comprises a lock bar slidably engaging with the second member of the hook and catch arrangement; the lock bar movable between a locked position locking the lid to a secure position to the base, and an unlocked position allowing the lid to be released from the base.
2. The container of claim 1 wherein the base and lid are each shaped to be a partial cylinder.
3. The container of claim 1, wherein the actuator comprises a knob having a squeeze actuator to release the knob from a fixed position to allow the knob to move and push on the lock bar to urge the lock bar from the locked position to unlocked position.
4. The container of claim 3 wherein the knob moves rotationally after released from the fixed position.
5. The container of claim 3 wherein the knob slides linearly after released from the fixed position.
6. The container of claim 3 wherein the knob includes a spring that returns the knob to the fixed position.
7. The container of claim 3 wherein: (a) the knob includes an internal ramp; and (b) the lock bar engages against the ramp as the knob is moved from the fixed position resulting in the lock bar sliding in a direction parallel to a longitudinal axis of the base to the unlocked position.
8. The container of claim 7 wherein the lock bar includes an end flange constructed and arranged to compress against the base, when the lock bar is moved to the unlocked position, and uncompress to return the lock bar to the locked position when the knob returns to the fixed position.
9. The container of claim 3 wherein: (a) the lock bar includes at least one notch corresponding to each hook and catch arrangement; and (b) when the lock bar is in the locked position, the notch is engaged with a projection on one of the first or second members of the hook and catch arrangement, and when the lock bar is in the unlocked position, the notch is free of engagement with the projection.
10. The container of claim 3 wherein the squeeze actuator includes a pair of opposite outwardly facing grip surfaces; and a pair of inwardly extending ribs.
11. The method of claim 3 wherein: (a) the container includes a plurality of subcavities and a

plurality of lids; one lid for each of the subcavities; and (b) the step of rotating the knob to move the lock bar from a locked position to an unlocked position controls locking and unlocking of all of the lids to the base.

12. A method of opening a container locked closed with a lock bar; the method comprising: (a) grasping a knob and moving the knob from a fixed position to a released position by squeezing a squeeze actuator on the knob; (b) rotating the knob to move the lock bar from a locked position to an unlocked position; and (c) while still grasping the knob, lifting a lid to release engagement of a secondary latch arrangement between the lid and a base of the container to expose an open cavity in the container.

13. The method of claim 12 further comprising releasing the knob and permitting the knob to spring back to the fixed position, and permitting the lock bar to spring back to the locked position.

14. The method of claim 12 further including: (a) moving the lid from an open position to a closed position by rotating the knob to move the lock bar from the locked position to the unlocked position; and (b) while still grasping the knob, moving the lid from the open position to the closed position.
